



LESSON 9.3

TRANSLATIONS

LESSON INTRODUCTION

MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

LEARNING TARGETS

- I can describe relationships between figures after a translation.
- I can perform a translation of a figure.
- I can perform a translation of a figure on the coordinate plane.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

ESSENTIAL QUESTIONS

- How are the preimage and image related after performing a translation?
- How do you translate a figure on and off the coordinate plane?

LESSON NARRATIVE

Students discover the characteristics of translations and how to perform them. Initially, students perform a one-dimensional translation (right/left or up/down). Then, students extend their skills to perform two-dimensional translations. Students also use prime notation and identify the coordinates of the preimage and image.

COMPONENT SUMMARY

9.3.1 WARM-UP (~5 MIN)

Describe transformations used to make pattern

9.3.3 GUIDED INSTRUCTION (15 MIN)

Perform translations on and off coordinate plane

9.3.5 WRAP-UP (~5 MIN)

Demonstrate understanding of performing and describing translations

9.3.2 EXPLORATION (10 MIN)

Identify characteristics of translations

9.3.4 TRY IT (10 MIN)

Practice performing translations on and off coordinate plane

RESOURCES

GLOSSARY

Key Terms:

- Translation

Additional Terms: N/A

MATERIALS

- (Optional) Graph Paper
- (Optional) Patty Paper

ADDITIONAL PREPARATION

- Enlarged visual representations of figures could be created on card stock and cut out prior to lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the preimage and image.
- Students may add/subtract the distance of the translation from the wrong coordinate.

THINGS TO CONSIDER

- Avoid forcing students to memorize a formula or relationship between the translation and the coordinates. The lesson scaffolds this so that students have the experience to determine the relationship.
- Consider having students use patty paper to copy the images to perform the transformations and/or determine if lengths and angles are congruent.

LITERACY AND LANGUAGE ROUTINES

Poll the Class

Consider polling the class for their answers to question 2b in 9.3.3 Guided Instruction and question 1a in 9.3.4 Try It. Ask students what their answers are for question 1a. Then, ask students to signal (i.e. raise hand or thumbs up) which answer they had. After polling students, have them pair up with a partner (ideally someone who has a different answer), each share how they obtained their answers, and resolve any different answers. This discussion will build student confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 9.3.2 Exploration, prompt to reflect on and discuss the characteristics they notice about translations. Ask students to consider how translations compare to reflections and rotations. Consider creating a Frayer Model for translations. Include the definition, visuals, examples, and non-examples for students to reference throughout the lesson.

DIFFERENTIATE INSTRUCTION

A kinesthetic entry-point can be offered by creating enlarged card stock figures and asking students to physically perform the translations in 9.3.2 Exploration, 9.3.3 Guided Instruction, and 9.3.4 Try It. Verbally describing the translations provides an auditory entry-point.

9.3.1 WARM-UP: DESCRIBING PATTERNS

TEACHER MOVES

Consider labeling each tile, for example I, II, III, IV (based on coordinate plane quadrants) to aid in communicating transformations. After having a variety of students share, discuss situations where there is more than one correct answer (for example a 90° rotation and a reflection).

DIFFERENTIATE INSTRUCTION

This activity provides an opportunity to review reflections and rotations. Consider providing students with a cutout of these tiles to physically manipulate. You can use a simplified pattern if reproducing the pattern on the tile is not feasible.



9.3.1 Warm-Up: Describing Patterns

A pattern is formed using ceramic tiles.



- Describe the geometric transformations that could have been used to make the pattern.
Answers will vary. The top left tile could be rotated 90° clockwise three times to create the other three tiles.

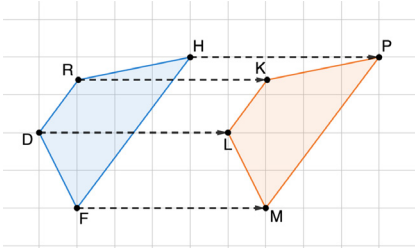


9.3.2 Exploration: Defining Translations

Work with your partner to complete the following.

The rigid transformation of “sliding” a figure is called a **translation**.

1. In the figure shown, quadrilateral $RDFH$ was translated to create quadrilateral $KLMP$.



- a. Each gridline in the image represents 1 centimeter (cm). Complete the table.

Corresponding Vertices	Distance (cm)
R and K	5 cm
D and L	5 cm
F and M	5 cm
H and P	5 cm

- b. What do you notice about the distances from the vertices of the preimage to the vertices of the image?
Answers will vary. I notice that the distances from the vertices of the preimage to the vertices of the image are all the same.

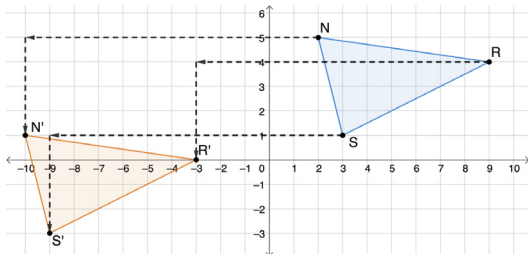
- c. Complete the statements.

Quadrilateral $RDFH$ was translated 5 cm to the

☐ left ☒ right to create quadrilateral $KLMP$. The mapping

lines for the translation are ☐ intersecting. ☒ parallel. ☐ perpendicular.

2. Rigid transformations can also be performed on a coordinate plane. The example shows a translation performed on $\triangle NRS$.



Triangle $N'R'S'$ is read as “n prime, r prime, s prime.” This notation is often used when a figure has been transformed. When prime notation is used, it is most commonly used for the vertices of the image.

The arrows show a possible path of movement. When a figure is translated on the coordinate plane, the figure’s movement is best described as a combination of horizontal and vertical movements, instead of diagonal.

9.3.2 EXPLORATION: DEFINING TRANSLATIONS

TEACHER MOVES

Translation is typically the easiest transformation for students to grasp. As you circulate listen for students who may be struggling. Also look for opportunities to build students who typically struggle but are showing an understanding of the content.

DIFFERENTIATE INSTRUCTION

Consider discussing the difference between parallel and perpendicular lines. Students may benefit from seeing a visual example to help them distinguish between these terms.

TEACHER MOVES

Students may struggle to distinguish between the preimage and image. Encourage students to label the corresponding points with prime notation before describing the translation. This will help students avoid making this common mistake.

9.3.2 EXPLORATION: DEFINING TRANSLATIONS (CONTINUED)

ENRICHMENT

- What do you notice about the distances and directions that each point was translated?
- What do you think would happen if one of the points was translated a different distance or direction than the other points?

a. Complete the statements to describe the translation of $\triangle NRS$ to $\triangle N'R'S'$.

- Point N was translated 12 units left and 4 units down to point N' .
- Point R was translated 12 units left and 4 units down to point R' .
- Point S was translated 12 units left and 4 units down to point S' .

b. What do you notice about the number of units and the direction that each point was translated?

I notice that each vertex on the preimage moves 12 units left and 4 units down to get to the image.

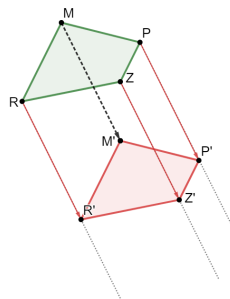


9.3.3 Guided Instruction: Translations On and Off the Coordinate Plane



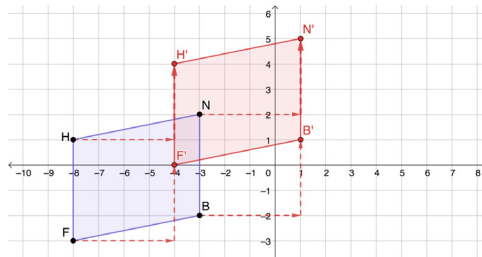
A **translation** is a transformation in which every point in a figure is moved in the same direction and by the same distance.

1. The figure shows quadrilateral $MRZP$ and the mapping for vertex M . Assume the transformation is a translation on the plane.



- a. Measure the length of the mapping from point M to point M' in cm. What is the distance?
4.7 cm
- b. Parallel line segments that can be used for the mapping are shown on the image above in grey. Use the measurement from Part A and the parallel line segments provided to complete the translation. Draw the mapping from vertices P , Z , and R that are the same length and parallel to the first mapping from point M to point M' . Label the image quadrilateral $M'R'Z'P'$.

2. Quadrilateral $BFHN$ is plotted on the coordinate plane.



- a. Translate quadrilateral $BFHN$ 4 units to the right and 3 units up. Label the image $B'F'H'N'$. Draw the mapping to show how each vertex moves from the preimage to the image.
- b. Record the coordinates of the vertices of the preimage and image.

	Coordinates		Coordinates
Point B	$(-3, -2)$	Point B'	$(1, 1)$
Point F	$(-8, -3)$	Point F'	$(-4, 0)$
Point H	$(-8, 1)$	Point H'	$(-4, 4)$
Point N	$(-3, 2)$	Point N'	$(1, 5)$

- c. Describe the relationship between the coordinates of the preimage and image.
Answers will vary. The x coordinate increases by 4 and the y coordinate increases by 3 from the preimage coordinates to the image coordinates. The image is now higher and farther right.

9.3.3 GUIDED INSTRUCTION: TRANSLATIONS ON AND OFF THE COORDINATE PLANE

TEACHER MOVES

Consider summarizing some of the positive things you heard during 9.3.2 Exploration. Also, consider addressing any common misconceptions using “my favorite mistake”. Consider creating a Frayer Model for translations. Include the definition, visuals, examples, and non-examples for students to reference throughout the lesson.

DIFFERENTIATE INSTRUCTION

Provide students with patty paper if they are struggling to perform the translation.

ENRICHMENT

- What is the relationship between corresponding points?
- How could you determine the coordinates of the image without performing the translation on the coordinate plane?

9.3.4 TRY IT: TRANSLATIONS ON AND OFF THE COORDINATE PLANE

QUESTIONING

Students may add/subtract the distance of the translation from the wrong coordinate. Remind students that the distance of the translation is added/subtracted to the x-coordinate for right/left translations and to the y-coordinate for up/down translations.

TEACHER MOVES

Poll the Class

Ask students what their answers are for question 1a. Then ask students to signal (i.e. raise hand or thumbs up) which answer they had. After polling students, have pair up with partner (ideally someone who has a different answer), each share how they obtained their answers, and resolve any different answers. This discussion will build student confidence and understanding.

DIFFERENTIATE INSTRUCTION

A kinesthetic entry-point can be offered by creating enlarged card stock figures and asking students to physically perform the translations or by using patty paper.

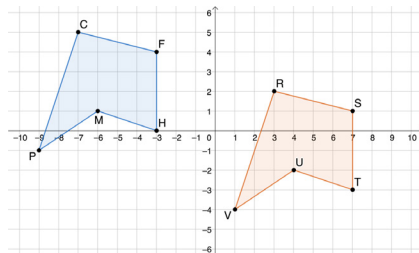
ENRICHMENT

- How was this mapping different than the translation in question 1?
- How can you distinguish between the preimage and image in question 3?



9.3.4 Try It: Translations On and Off the Coordinate Plane

1. In the figure below, polygon $RSTUV$ is the result of a translation performed on polygon $CFHMP$.



- a. Record the coordinates of the preimage and the image.

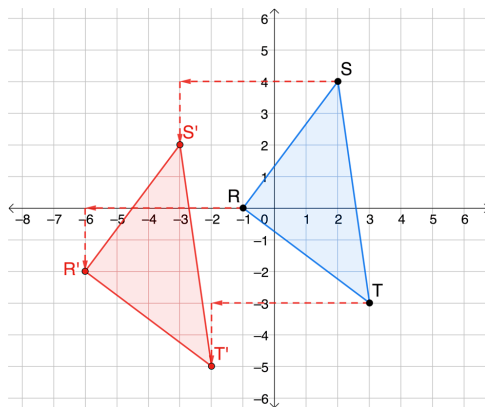
Vertex	Coordinates	Vertex	Coordinates
Point C	$(-7, 5)$	Point R	$(3, 2)$
Point F	$(-3, 4)$	Point S	$(7, 1)$
Point H	$(-3, 0)$	Point T	$(7, -3)$
Point M	$(-6, 1)$	Point U	$(4, -2)$
Point P	$(-9, -1)$	Point V	$(1, -4)$

- b. Complete the statement.

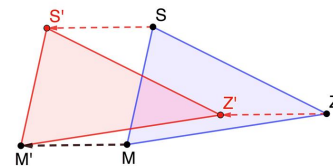
Polygon $CFHMP$ was translated 10 units

☐ right and ☐ left and 3 units ☐ up and ☐ down to create polygon $RSTUV$.

2. Translate $\triangle RST$ 5 units left and 2 units down to form $\triangle R'S'T'$. Label the image and draw the mapping.



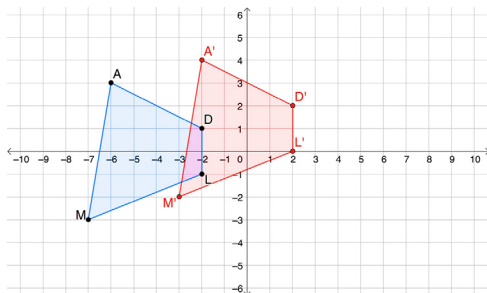
3. Use a straightedge to complete the translation of $\triangle MSZ$ started below. Label the image $\triangle M'S'Z'$, and draw the mapping of vertices S and Z .





9.3.5 Wrap-Up: Ticket Out the Door

1. Quadrilateral $ADLM$ is shown on the coordinate grid.
 - a. Translate quadrilateral $ADLM$ 4 units right and 1 unit up to create the image quadrilateral $A'D'L'M'$. Label the image.



- b. Describe at least two characteristics of the relationship between the two figures after the translation.
Answers will vary. The two figures have corresponding sides that are the same length and corresponding angles that have the same measure.

9.3.5 WRAP-UP: TICKET OUT THE DOOR

QUESTIONING

If students finish early, ask them to write a reflection on how translations compare to reflections and rotations.